Shren Patel

Senior Backend Engineer — AI Agent Integrations

patel.shrenil@gmail.com · +1 613-551-9989 · Canada remote · NA-friendly · EST/PST
linkedin.com/in/shrenp · github.com/blopit · shrenil.com

SUMMARY

10+ years backend engineering, third-party enterprise integrations. Built CentrOS MCP-style tool registry + AI-to-human handoff: scoped OAuth, JSON-schema validation, idempotent retries, audit logs, session lifecycle; 0 lost context across 100% of escalation events; 3,980+ Vitest tests on main. Reliability seam agent automation needs in enterprise ecosystems.

EXPERIENCE

Founder CentrOS — AI Agent Orchestration Platform *Feb 2026 – Present*
Remote · Canada • TypeScript · Hono · PostgreSQL · Drizzle · Vitest · LLM tool-calling

- Built CentrOS tool/integration layer: standardized API contracts, scoped OAuth, schema validation, idempotent retries, side-effect logging, audit trails — every agent-invoked external action is 1 typed call with full provenance.
- Designed agent runtime on LangGraph state graphs + Temporal-style durability: durable execution, checkpointing, crash recovery, retries, versioned rollouts — 0 silent run loss across 6 months.
- Designed AI-to-human handoff with 100% context preservation: shared state contract, event history, durable IDs, operator-readable summaries, ownership transitions surviving interruption — 0 lost context in escalations.
- Dogfooded CentrOS on 23-unit motel vertical (guest comms, refunds, vendor invoices, compliance) — 3,980+ Vitest tests across 347 files passing on main; CI-backed live ops.

Vertical automation & ops dogfood Conscious Creations *Sep 2022 – Present*
Remote · Canada • Python · FastAPI · LLM tool-calling · workflow automation

- Designed AI-to-human handoff: shared state contract, full conversation history, durable IDs, operator-readable summaries — 0 lost context across 100% of escalation events in live ops.
- Prototyped accommodations workflows on CentrOS: 3-stage guest-triage pipeline (classify → plan → draft) in Python/FastAPI with HITL approval gates before any irreversible action.

Creator, WEBTOON Original — AXED WEBTOON *Jun 2018 – Dec 2020*
Remote • Python · OpenCV · image processing

- Designed queue → tooling → review → ship operator workflow at 2M+ monthly views on WEBTOON Original — precursor to today's HITL agent + human production system at CentrOS.

Senior Frontend Developer Faire *Jun 2021 – Aug 2022*
Remote · Canada • React · TypeScript · MobX · Jest · Storybook · A/B testing

- Drove site-wide perf gains via A/B testing + staged rollouts; same guardrail discipline now applied to agent versioning, shadow traffic, and canary releases at CentrOS — 5+ wins shipped.

SKILLS

Integrations & Real-Time Systems:

- REST · gRPC · webhooks · OAuth · third-party enterprise APIs
- Context propagation · session lifecycle · AI-to-human handoff · state contracts
- Idempotency · retries · timeouts · audit logs · rollback · graceful degradation
- Real-time / streaming architectures · low-latency orchestration · async patterns
- Observability: handoff statistics · context-transfer quality · API failure patterns

Distributed Systems & Infra:

- Distributed systems · scalable backend services · API design
- Kubernetes · Docker · AWS · GCP · serverless · CI/CD

Languages:

- Python · TypeScript · Node.js · Go (familiar)

EDUCATION

University of Waterloo Bachelor of Software Engineering (BSE) — Computer Software Engineering *2012 – 2018*
Waterloo, ON